

INTRODUCTION TO APPLIED NUMERICAL METHODS

Taylor Series, Truncation Error & Error Propagation

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TAYLOR SERIES: AN EXAMPLE

Use 0th through 4th Taylor Series expansions to approximate

$$f(x) = -0.1x^4 - 0.15x^3 - 0.5x^2 - 0.25x + 1.2$$

Computing function's value at $x_{i+1} = 1$ by expanding the series at $x_i = 0$

Solution:

Query point: $x_{i+1} = 1$; Expansion Point: $x_i = 0$; step size $h = 1$

$$f(x_{i+1}) = f(x_i) + f'(x_i)h + \frac{f''(x_i)}{2!}h^2 + \frac{f^{(3)}(x_i)}{3!}h^3 + \dots + \frac{f^{(n)}(x_i)}{n!}h^n + R_n$$

$$\text{true value: } f(1) = 0.2$$

$$0^{\text{th}} \text{ Order (n = 0): } f(x_{i+1}) = f(1) \approx f(0) = 1.2 \quad \text{true err: } E_t = 0.2 - 1.2 = -1.0$$

$$1^{\text{st}} \text{ Order (n = 1): } f'(x) = -0.4x^3 - 0.45x^2 - 1.0x - 0.25 \Rightarrow f'(0) = -0.25$$

$$f(x_{i+1}) = f(1) \approx f(0) - 0.25h = 0.95$$

$$E_t = 0.2 - 0.95 = -0.75$$

TAYLOR SERIES: AN EXAMPLE

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$$\text{true value: } f(1) = 0.2$$

$$f'(x) = -0.4x^3 - 0.45x^2 - 1.0x - 0.25$$

$$2^{\text{nd}} \text{ Order } (n = 2): \quad f''(x) = -1.2x^2 - 0.9x - 1.0 \quad \Rightarrow \quad f''(0) = -1.0$$

$$f(x_{i+1}) = f(1) \approx f(0) - 0.25h - \frac{1}{2!}h^2 = 0.45$$

$$E_t = 0.2 - 0.45 = -0.25$$

TAYLOR SERIES: AN EXAMPLE

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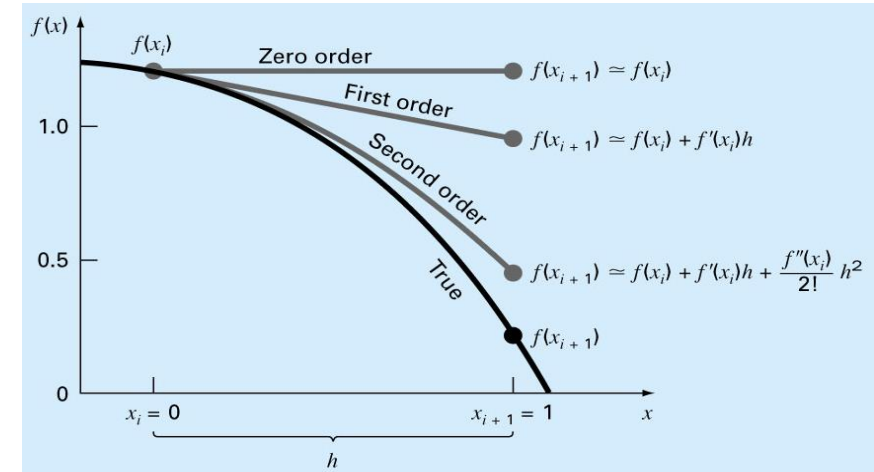
$$f(x_{i+1}) = f(x_i) + f'(x_i)h + \frac{f''(x_i)}{2!}h^2 + \frac{f^{(3)}(x_i)}{3!}h^3 + \dots + \frac{f^{(n)}(x_i)}{n!}h^n + R_n$$

4th Order ($n = 4$): $f^{(4)}(0) = -2.4$

$$f(x_{i+1}) = f(1) \approx f(0) - 0.25h - \frac{1}{2}h^2 - \frac{0.9}{3!}h^3 - \frac{2.4}{4!}h^4 = 0.2$$

$$E_t = 0$$

$$R_4 = O(h^5) = 0$$



true value: $f(1) = 0.2$

TAYLOR SERIES: AN EXAMPLE

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

- Technical Findings
 - The n^{th} -order Taylor series expansion will be exact for an n^{th} -order polynomial (derivatives of higher order are all zeros and $R_n = 0$)
 - For general functions (differentiable and continuous), such as exponentials (exp) and sinusoids (sin, cos), a finite number of terms will never produce an exact estimate.
 - In practice, the inclusion of only a few terms could yield an approximation close enough to the true value
 - We can also use the step size h to control the truncation error

Remainder $R_n = O(h^{n+1})$

$R_0 = O(h) \sim h$	Halving step size will halve the error
$R_1 = O(h^2) \sim h^2$	Halving step size will quarter the error

R_n is very useful to evaluate the **comparative error** of numerical methods based on Taylor series expansions

NUMERICAL DIFFERENTIATION

- In many engineering problems, we need to numerically approximate the derivative
 - The derivative of function's values with respect to independent variables, e.g., time t and spatial coordinates (x, y, z)
 - Derived from Taylor series expansion

1st Order Taylor series ($n = 1$) expanded @ x_i :

$$f(x_{i+1}) = f(x_i) + f'(x_i)h + O(h^2)$$

$$f'(x_i)h = f(x_{i+1}) - f(x_i) - O(h^2)$$

$$f'(x_i) = \frac{f(x_{i+1}) - f(x_i)}{h} - O(h)$$

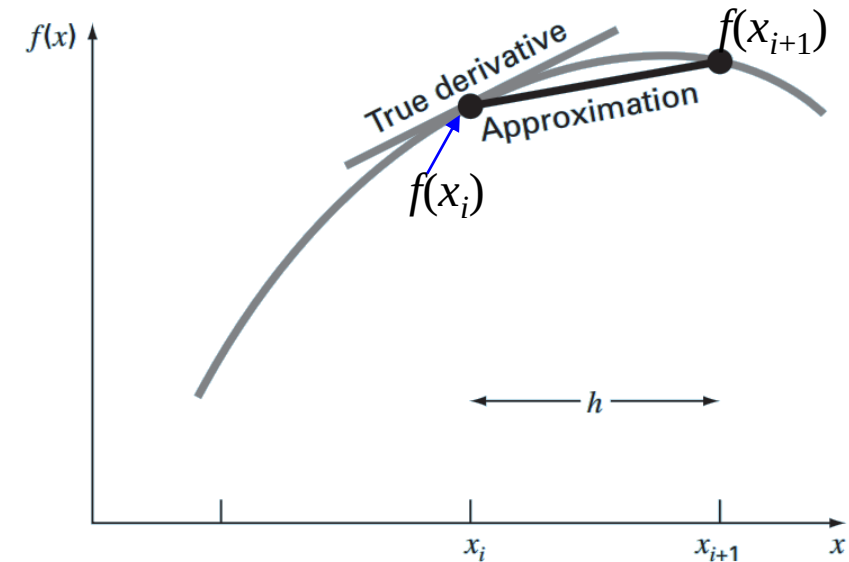
h ← Step size

Truncation error of order h

$$f'(x_i) = \frac{\Delta f_i}{h} - O(h)$$

First forward difference

First finite divided difference



“forward” difference because it uses data at i and $i+1$ to estimate the derivative at $f(x_i)$

BACKWARD DIFFERENCE APPROXIMATION OF THE 1ST DERIVATIVE

1st Order Taylor series ($n = 1$) expanded @ x_i :

$$f(x_{i-1}) = f(x_i) - f'(x_i)h + O(h^2)$$

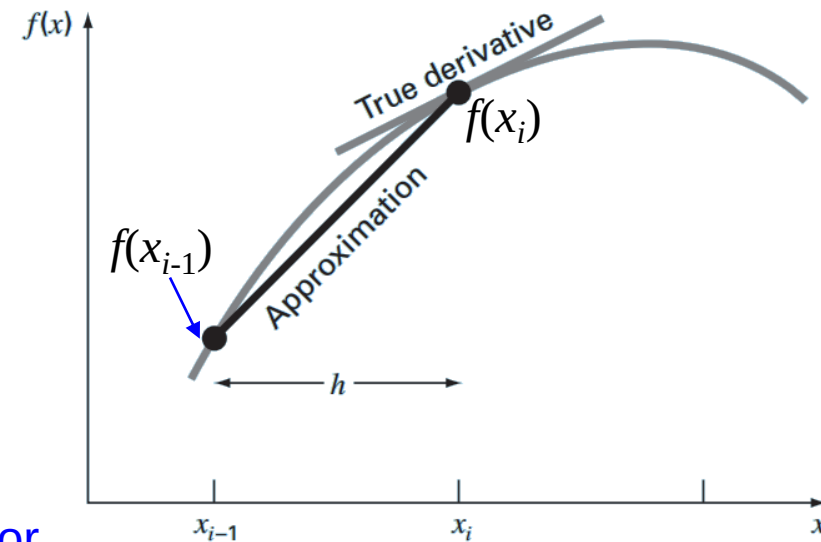
$$f'(x_i)h = f(x_i) - f(x_{i-1}) + O(h^2)$$

$$f'(x_i) = \frac{f(x_i) - f(x_{i-1})}{h} + O(h)$$

Step size

$$f'(x_i) = \frac{\nabla f_i}{h} + O(h)$$

First backward
difference



Trunc. error
of order h

“backward” difference because it uses data at i and $i-1$ to estimate the derivative at $f(x_i)$

CENTERED DIFFERENCE APPROXIMATION OF THE 1ST DERIVATIVE

2nd Order Taylor series (n = 2) expanded
@ x_i :

$$f(x_{i+1}) = f(x_i) + f'(x_i)h + \frac{f''(x_i)}{2!}h^2 + O(h^3)$$

2nd Order Taylor series (n = 2) expanded
@ x_i :

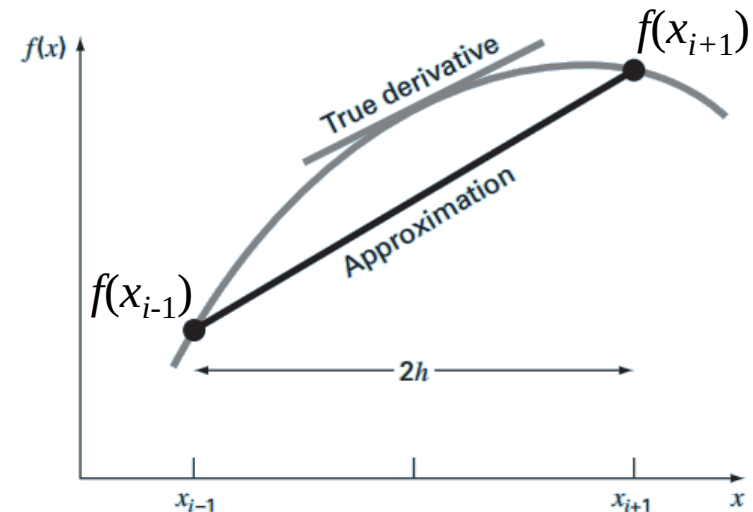
$$f(x_{i-1}) = f(x_i) - f'(x_i)h + \frac{f''(x_i)}{2!}h^2 - O(h^3)$$

minus “-”

$$f(x_{i+1}) - f(x_{i-1}) = 2f'(x_i)h + 2O(h^3)$$

$$f'(x_i) = \frac{f(x_{i+1}) - f(x_{i-1}))}{2h} - O(h^2)$$

Of order h^2



“centered” difference because it uses data at $i+1$ and $i-1$ to estimate the derivative at $f(x_i)$

AN EXAMPLE

Use forward and backward difference and a centered difference to estimate the first derivative of

$$f(x) = -0.1x^4 - 0.15x^3 - 0.5x^2 - 0.25x + 1.2 \quad \leftarrow \text{Error (1.25) in textbook}$$

at $x = 0.5$ using a step size $h = 0.5$. Repeat the computation using another step size $h = 0.25$.

Note that the analytic solution of the 1st derivative is given by

$$f'(x) = -0.4x^3 - 0.45x^2 - x - 0.25$$

Solution:

True value $f'(0.5) = -0.9125$

For $h = 0.5$, the function values to be used

$$x_{i-1} = 0 \quad f(x_{i-1}) = 1.2$$

$$x_i = 0.5 \quad f(x_i) = 0.925$$

$$x_{i+1} = 1.0 \quad f(x_{i+1}) = 0.2$$

Forward divided difference

$$f'(0.5) = \frac{f(x_{i+1}) - f(x_i)}{h} = \frac{0.2 - 0.925}{0.5} = -1.45, \quad \varepsilon_t = \left| \frac{-0.9125 + 1.45}{-0.9125} \right| = 58.9\%$$

AN EXAMPLE

Solution:

True value $f'(0.5) = -0.9125$

For $h = 0.5$, the function values to be used

$$x_{i-1} = 0 \quad f(x_{i-1}) = 1.2$$

$$x_i = 0.5 \quad f(x_i) = 0.925$$

$$x_{i+1} = 1.0 \quad f(x_{i+1}) = 0.2$$

Backward divided difference

$$f'(0.5) = \frac{f(x_i) - f(x_{i-1}))}{h} = \frac{0.925 - 1.2}{0.5} = -0.55, \quad \varepsilon_t = \left| \frac{-0.9125 + 0.55}{-0.9125} \right| = 39.7\%$$

centered divided difference

$$f'(0.5) = \frac{f(x_{i+1}) - f(x_{i-1}))}{2h} = \frac{0.2 - 1.2}{1.0} = -1, \quad \varepsilon_t = \left| \frac{-0.9125 + 1}{-0.9125} \right| = 9.6\%$$

AN EXAMPLE

Solution:

$$\begin{array}{lll} \text{For } h = 0.25 & x_{i-1} = 0.25 & f(x_{i-1}) = 1.10351563 \\ & x_i = 0.5 & f(x_i) = 0.925 \\ & x_{i+1} = 0.75 & f(x_{i+1}) = 0.63632813 \end{array}$$

Forward divided difference

$$f'(0.5) = \frac{f(x_{i+1}) - f(x_i)}{h} = \frac{0.63632813 - 0.925}{0.25} = -1.155, \quad \varepsilon_t = \left| \frac{-0.9125 + 1.155}{-0.9125} \right| = 26.6\%$$

Backward divided difference

$$f'(0.5) = \frac{f(x_i) - f(x_{i-1})}{h} = \frac{0.925 - 1.10351563}{0.25} = -0.714, \quad \varepsilon_t = \left| \frac{-0.9125 + 0.714}{-0.9125} \right| = 21.7\%$$

centered divided difference

$$f'(0.5) = \frac{f(x_{i+1}) - f(x_{i-1}))}{2h} = \frac{0.63632813 - 1.10351563}{0.5} = -0.934, \quad \varepsilon_t = \left| \frac{-0.9125 + 0.934}{-0.9125} \right| = 2.4\%$$

STABILITY AND CONDITION

What is “condition”?

It relates to sensitivity of the functional values $f(x)$ to changes in its input values x .

“Numerically unstable”

- The uncertainty (error) of the input values is grossly magnified by the function (or the numerical method).

First-order Taylor series

$$f(x) = f(\tilde{x}) + f'(\tilde{x})(x - \tilde{x}) \quad \rightarrow \quad \frac{f(x) - f(\tilde{x})}{f(x)} \cong \frac{f'(\tilde{x})(x - \tilde{x})}{f(\tilde{x})} \quad \textcircled{1} \quad \text{Relative error of } f(x) \text{ caused by approximating } x \text{ with } \tilde{x}$$

$$\frac{x - \tilde{x}}{x} \cong \frac{x - \tilde{x}}{\tilde{x}} \quad \textcircled{2} \quad \text{Relative error of } x \text{ caused by approximating } x \text{ with } \tilde{x}$$

A **condition number** can be defined as the ratio of the two relative errors, i.e., relative error of $f(x)$ to the relative error of x

$$\text{Condition number } C = \frac{\textcircled{1}}{\textcircled{2}} = \left| \frac{\tilde{x} \cdot f'(\tilde{x})}{f(\tilde{x})} \right|$$

$C < 1$ error is attenuated
 $C = 1$ relative error is the same
 $C > 1$ error is amplified
 $C \gg 1$ error is significantly amplified (**ill-conditioned**)

AN EXAMPLE

Compute and interpret the condition number of function $f(x) = \tan x$

for $\tilde{x} = \frac{\pi}{2} + 0.1\left(\frac{\pi}{2}\right)$ and $\tilde{x} = \frac{\pi}{2} + 0.01\left(\frac{\pi}{2}\right)$

Solution:

$$\text{Condition number } C = \left| \frac{\tilde{x} \cdot f'(\tilde{x})}{f(\tilde{x})} \right|$$

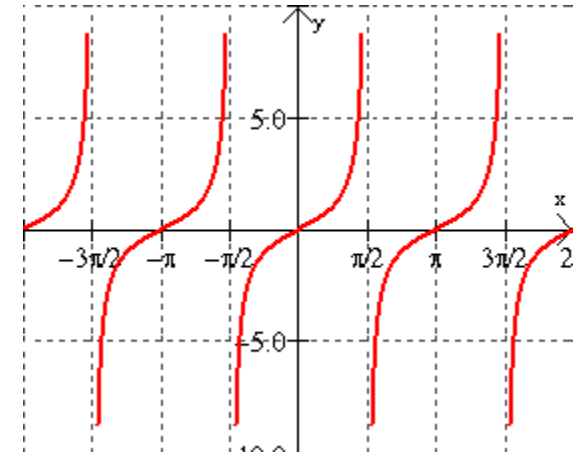
$$f'(x) = \left(\frac{\sin x}{\cos x} \right)' = \frac{\partial \left(\frac{\sin x}{\cos x} \right)}{\partial x}$$

$$\text{Quotient Rule: } \left(\frac{f}{g} \right)' = \frac{f' \cdot g - f \cdot g'}{g^2}$$

$$f'(x) = \frac{(\sin x)' \cos x - \sin x (\cos x)'}{\cos^2 x} = \frac{\cos x \cdot \cos x + \sin x \cdot \sin x}{\cos^2 x} = \frac{1}{\cos^2 x}$$

$$\text{Condition number} = \left| \frac{\tilde{x} (1/\cos^2 \tilde{x})}{\tan \tilde{x}} \right|$$

AN EXAMPLE



for $\tilde{x} = \pi/2 + 0.1(\pi/2)$

$$\tan(\tilde{x}) = -6.3138 \quad \cos(\tilde{x}) = -0.1564$$

$$\text{Condition number} = \left| \frac{\tilde{x}(1/\cos^2 \tilde{x})}{\tan \tilde{x}} \right| = \left| \frac{1.7279}{-6.3138 \times (-0.1564)^2} \right| = \left| \frac{1.7279 \times (40.88)}{-6.3138} \right| = 11.1876$$

for $\tilde{x} = \pi/2 + 0.01(\pi/2)$

$$\tan(\tilde{x}) = -63.6567 \quad \cos(\tilde{x}) = -0.0157$$

$$\text{Condition number} = \left| \frac{\tilde{x}(1/\cos^2 \tilde{x})}{\tan \tilde{x}} \right| = \left| \frac{1.5865}{-63.6567 \times (-0.0157)^2} \right| = \left| \frac{1.5865 \times (4057)}{-63.6567} \right| = 101.1116$$

ill-conditioning